

DATA ANALYTICS PROJECT REPORT

Project Title: Sentiment Dataset Analysis Using Python

1. Introduction

Social media sentiment analysis helps organizations understand public opinion by analyzing textual data. This project demonstrates a complete data analytics workflow, including data cleaning, exploratory data analysis (EDA), and data visualization using Python.

2. Objectives

- Clean and preprocess the dataset.
- Handle missing values and duplicate records.
- Standardize inconsistent data formats.
- Perform exploratory data analysis.
- Create visualizations to identify trends and relationships.

3. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, Seaborn, and Visual Studio Code.

4. Dataset Description

The dataset contains social media posts with attributes such as Text, Sentiment, Timestamp, Platform, Likes, Retweets, Country, Year, Month, Day, and Hour.

5. Task 1 - Data Cleaning and Processing

The dataset was loaded using Pandas. Missing values were identified using `isnull()`. Duplicate records were removed using `drop_duplicates()`. Numerical missing values were filled using the mean, while categorical missing values were filled using the mode where applicable. Text fields were standardized by removing extra spaces and converting sentiment labels to uppercase. Timestamp values were converted into datetime format. Finally, the cleaned dataset was saved as *cleaned_sentiment_dataset.csv*.

6. Task 2 - Exploratory Data Analysis

Summary statistics including mean, median, mode, and standard deviation were calculated. Correlation analysis was performed on numerical variables. Histograms, box plots, scatter plots, and a correlation heatmap were generated to understand data distribution and relationships.

7. Task 3 - Basic Data Visualization

The following visualizations were created:

- Bar Chart - Sentiment Distribution
- Bar Chart - Platform Distribution
- Line Chart - Average Likes by Month
- Scatter Plot - Likes vs Retweets

Each chart includes appropriate titles, axis labels, and was exported as an image for reporting.

8. Results

The project produced a clean and consistent dataset suitable for further analysis. The visualizations highlighted sentiment distribution, user engagement trends, and relationships between likes and retweets.

9. Conclusion

This project successfully implemented the core stages of a data analytics workflow. Data cleaning improved data quality, EDA revealed useful insights, and visualizations communicated the findings effectively. The cleaned dataset can be used for future predictive analytics or machine learning applications.

10. References

Python Documentation, Pandas Documentation, Matplotlib Documentation, Seaborn Documentation.